

DWI Prompt Optimizer

Management Brief, Current Evidence, and Future Direction

Plain-language purpose: what works today, where the limits are, and how DWI could later support responsible AI adoption

Evidence date: 1 September 2026

Audience: business, engineering, delivery, operations, and AI leadership

Product evidence version: 5088dbf6e5fdeea2415b991c429de8e5229d7a7b

Status: local proof of concept; future ideas are clearly marked and are not built yet

Important boundary. This document does not claim production readiness, live external-AI output quality, release approval, project-management integration, or a deployed management reporting system.

Executive summary

DWI helps developers give AI the right project background before asking it to do work. After login, it gathers a limited set of project facts with the developer's consent, shows those facts for review and correction, and saves only the approved project summary. Prompt Optimizer then reuses that approved knowledge for later tasks. It is more than a generic prompt editor.

The current local proof of concept provides a consistent journey: review approved project facts, describe the work, see sources and assumptions, identify important missing information, prepare a local draft, review it, save it, or reset the current task without deleting approved project knowledge. Engineering checks passed on the recorded implementation version, including 421 tests, type checks, packaging, local installed checks, narrow-screen review, and successful continuous-integration checks.

Three limits must remain clear:

- DWI can show an important unanswered question, but the guided one-question-at-a-time answer step is not finished.
- The latest automated restart check could not run because macOS automation could not reach the consent window. This is not evidence of a product failure, but it is also not a new pass. Earlier restart evidence remains available.
- Phase 4 testing of result quality from a live external AI service remains blocked because the agreed 12,000-token usage limit was exceeded. No retry was made.

The future opportunity is to connect approved task information from a project-management system to DWI. Agreed rules could turn task priority, risk, and complexity into a clear AI usage limit. DWI's internal prompt-building engine could use that limit when deciding how much approved context to include. Separately, outcome-only feedback—such as helpful or not helpful, corrected, saved, reset, time taken, and limit adherence—could be grouped for management insight without copying prompts, generated output, code, project content, file paths, or employee names.

These future features are not built today. A usage limit would never authorize spending or external AI use, and feedback would improve tools, training, and delivery practices—not rank or monitor individual employees.

Recommendation

Demonstrate the current proof of concept with representative work, finish the guided clarification and restart checks, agree the privacy and spending guardrails, run the proposed allocation in dry-run mode, and only then consider one small controlled pilot.

1. The management challenge

AI use can spread quickly while each team develops its own habits. Without a shared approach, the organization may see more activity without knowing whether work is becoming faster, clearer, safer, or easier to review.

Four common problems

Problem	What it means in everyday work	Why leaders should care
Repeated setup	Developers explain the same project background again for each task and tool	Time is lost and the same facts are described differently
Uneven results	Important rules, risks, and assumptions may be missing	Polished-looking work can still create review effort or rework
Unclear AI effort	Priority and complexity are not linked to an agreed usage limit	AI capacity and cost are harder to plan and explain
Little useful feedback	Usage counts show activity, not whether AI actually helped	Leaders cannot target training, tools, or process improvements confidently

The opportunity

DWI offers one clear way of working from approved project knowledge to a reviewed result. The aim is not to remove human judgment. The aim is to give developers a better starting point, make assumptions visible, and keep decisions with the developer. This creates a possible foundation for responsible growth later: project priority can inform a clear limit, work remains reviewable, and privacy-safe outcome information can help improve the system.

2. What DWI does today

Start with project knowledge the developer approves

DWI begins after login with a limited project check. The developer is told what will be inspected and gives consent. DWI prepares a short project summary, and the developer can review, correct, and approve it.

That approved project summary is saved for later work. In the product it is called **Project Meta Context**. In plain terms, it is the agreed set of project facts that later tasks may reuse.

The current journey

Step	What the developer does	What DWI contributes
1. Review project facts	Checks and approves the project summary	Reuses bounded project knowledge instead of starting from nothing
2. Describe the task	Enters the task, expected result, and importance	Gives each AI-assisted task a consistent starting form
3. See what is known	Reviews sources, assumptions, and missing information	Shows where information came from and where an answer is still needed
4. Prepare a local draft	Moves through the local preparation steps	Produces a rule-based draft without requiring an external AI service
5. Review and decide	Corrects, saves, copies, or rejects the work	Keeps the final decision with the developer
6. Reset one task	Clears the current task when needed	Keeps the approved project facts for the next task

What works today and what does not

Available in the local proof of concept	Not available or not proven today
Approved project summary	Production-wide rollout
Clear task and importance fields	Finished guided question-and-answer step
Visible sources and assumptions	Proven quality improvement from a live external AI service
Important missing information shown	Project-management system connection
Local rule-based draft and review	Automatic task-based AI usage limits
Saved work and safe task reset	Management dashboards or group reporting

3. Benefits for people and delivery

Benefits visible in the local proof of concept

- **Less setup time:** approved project knowledge can be reused instead of explained again.
- **Fewer missed requirements:** known rules and constraints stay visible while work is prepared.
- **Less mental effort:** developers spend less time searching, repeating, and tracking information across tools.
- **Easier review:** sources, assumptions, and decisions are simpler to check.
- **Shared understanding:** teams can begin from the same approved project facts.
- **Developer control:** the developer reviews, corrects, saves, or resets the work.

Value at each level

Level	Main benefit	Decision supported
Developer	Clearer day-to-day work with less repeated setup	Which support helps this task
Team lead	More consistent handoffs and fewer recurring knowledge gaps	Which shared knowledge or team practice needs improvement
Software delivery operations	Better view of delays, fallback use, and process friction	Which delivery or platform problem to fix first
Management and AI Center of Excellence	Better view of useful adoption, training needs, and overall capacity	Where to invest, expand, constrain, or stop

Why this is different from measuring usage alone

More prompts or more AI activity do not automatically mean more value. High activity may indicate repeated attempts, confusion, or poor results. Management should look at usefulness, correction effort, time to a usable draft, safe fallback, and adherence to agreed limits—not just volume.

4. What remains unfinished

4.1 The guided question step

DWI can identify important missing information and show a material question. The next user-facing step is not complete: ask one short question, let the developer answer or skip it, apply the answer only to the current task, and avoid using an old answer after the task or project facts change.

This should be finished before future project-management data is allowed to change how much context DWI uses.

4.2 The latest restart check

Earlier testing showed that approved project facts and saved task work could survive a restart, and that resetting one task did not remove approved project knowledge.

The latest repeat of this check could not begin because macOS automation could not see the consent window in the temporary test copy of VS Code. Three attempts stopped before testing the product flow. This is a test-automation problem, not proof of a product failure, but it is also not a new pass for the latest version.

4.3 Live external-AI output quality

The local product and its safety checks do not prove that an external AI service improves the result. The one approved readiness call used 14,531 AI usage units, often called **tokens**, against an agreed limit of 12,000.

The process stopped safely. No quality test followed, no comparison benchmark ran, and no retry was made. A future attempt would need separate approval covering the external service, information that may be sent, retention, maximum calls, and cost.

4.4 Production and release

The proof of concept is not a production release. It has not been merged to the main branch, tagged as a baseline, released through a marketplace, or approved for broad rollout.

5. Future direction: project-management data, clear AI limits, and safe feedback

These ideas describe a possible next stage. They are not built into the current product.

5.1 Optional project-management connection

A future connection to a project-management system could provide a small approved set of task details:

- task or work-item reference;
- delivery period or sprint;
- type of work;
- priority;
- risk or importance;
- broad complexity band;
- delivery stage;
- version of the approved company rule.

DWI should still work when this connection is unavailable. The project-management system should not become a required dependency for the basic local journey.

5.2 Clear AI usage limit

Agreed company rules could turn approved task details into a clear usage limit. This is sometimes called a **token budget**, because tokens are the units used to measure AI input and output.

The limit could help DWI decide how much approved project context to include and how deeply to prepare the task. The order of control should remain simple:

Consent and privacy → hard safety limits → company rules → task limit → DWI prepares the work → explicit external-AI action → developer review

The usage limit must never:

- approve spending;
- approve external AI use;
- raise a hard safety limit;
- send information without a clear user action;
- replace developer review.

5.3 Outcome-only feedback

Future feedback could record simple outcomes without copying the work itself:

- helpful, mixed, or not helpful;
- saved, reused, corrected, reset, or abandoned;
- type of correction, without the text;
- missing question answered or skipped;
- local fallback used;
- broad time band;
- estimated and observed AI usage where available;
- stayed within the agreed limit or stopped safely.

This feedback could help improve guidance, training, budgets, and delivery processes. It must not copy prompts, generated results, code, project content, file paths, free-text notes, or employee names.

5.4 The future improvement loop

Approved task details → agreed company rule → clear AI usage limit → DWI prepares work → developer decision → outcome-only feedback → group-level improvement

The loop should advise people. It should not automatically change safety rules or score an employee.

6. Management goals and useful measures

Management goal	Useful information	Decision it supports
Useful AI adoption	Helpful outcomes, save or reuse, correction type, workflow coverage	Where AI helps and where training or redesign is needed
Predictable AI capacity and cost	Usage limit, observed use, limit adherence, time band	Capacity planning, budget bands, and exception rules
Better delivery quality	Missing-information rate, correction patterns, rework, fallback	Which workflow stages need better context or checks
More consistent knowledge	Reuse of approved project facts, recurring missing-information categories	Which project knowledge or guidance needs improvement
Responsible control	Consent state, rule version, safe-stop reason, explicit action	Whether safeguards are working as intended
Better developer experience	Usefulness, correction burden, reset or abandon outcome	Tool, training, and support priorities

Recommended measures for a small pilot

- use of approved project knowledge;
- time to a developer-usable first draft;
- important missing information found and resolved;
- accepted, corrected, abandoned, or reset outcome;
- avoidable rework category;
- fallback and failure rate;
- difference between estimated and observed AI usage;
- percentage of work staying within the agreed limit;
- usefulness by workflow or group;
- opt-out and deletion requests handled correctly.

Each measure should answer a real decision. If no decision would change, the organization should not collect the data.

7. Privacy, employee trust, and spending safeguards

Non-negotiable rules

- DWI continues to work without a project-management or reporting service.
- The developer approves project facts before they are reused.
- Moving through the local steps does not call an external AI service.
- An AI usage limit is a cap, not permission to spend or transmit information.
- Hard privacy and safety limits always take priority over task urgency.
- The developer reviews the result and decides what to keep.
- Raw prompts, generated results, code, project files, paths, and free-text notes stay out of management reporting.
- Reports are grouped by workflow or a sufficiently large group, not by named person.
- Individual ranking, performance scoring, and disciplinary use are prohibited.
- People can understand the data collected, its purpose, retention, access, and deletion rules.

Information handling in plain language

Information	Example	Default handling
Approved project facts	Reviewed project summary	Kept with the local project; not copied into management reporting
Task content	Prompt, source text, generated draft	Kept local to the task; excluded from reusable reporting
Optional written feedback	Developer’s free-text note	Local unless the developer separately chooses to share it
Outcome-only feedback	Helpful, saved, corrected, time band	May be grouped only under a reviewed privacy agreement
Project-management details	Priority, risk, delivery stage	Keep to the minimum fields needed for the agreed rule
Company rule	Version, limit, reason	Recorded so the decision can be explained and rolled back

Employee-trust rule

The feedback loop exists to improve the product, training, project knowledge, and software delivery. It is not a measure of individual developer productivity.

8. Recommended path from demo to small pilot

Stage 1 — Try the current local demo

- use representative developer work;
- look for clearer tasks, reused project facts, fewer omissions, and easier review;
- collect local qualitative feedback without centralizing raw content;
- decide whether the idea is useful enough to continue.

Stage 2 — Finish the core experience

- complete the guided one-question answer or skip step;
- ensure an old answer cannot affect a changed task;
- repeat the restart, resume, and reset journey on the same version;
- record the exact product version and result.

Stage 3 — Agree the guardrails

- choose one target workflow;
- define the minimum project-management fields;
- agree the AI usage limits, priority order, exceptions, and stop switch;
- agree what outcome information may be collected;
- agree retention, access, grouping, deletion, and opt-out rules;
- choose success and stop measures before the pilot.

Stage 4 — Run a dry-run

- calculate suggested limits without changing real work;
- test missing or old project-management data;
- test urgent tasks that conflict with a hard limit;
- confirm the dry-run cannot approve spending or call an external AI service;
- check that every suggested limit is understandable.

Stage 5 — Consider one small controlled pilot

- use one workflow, one group, one rule version, one time period, and one agreed maximum;
- keep external AI use explicit and human review mandatory;
- stop on a privacy, safety, or spending-rule breach;
- decide whether to improve, expand, constrain, or stop.

9. Management decision

Decision requested now

Experience the current local proof of concept and decide whether it is useful enough to justify completing the core experience and designing a small dry-run.

Decisions not requested now

- production rollout;
- release or marketplace publication;
- live external-AI quality testing;
- permission to spend or transmit project information;
- project-management system integration;
- management reporting deployment;
- employee monitoring;
- merge to the main branch or product baseline approval.

Questions for the meeting

1. Does starting from approved project knowledge solve a meaningful delivery problem?
2. Which one workflow would provide the clearest, lowest-risk demonstration?
3. Which outcomes would prove usefulness beyond simple activity counts?
4. Which privacy, employee-trust, and spending rules are non-negotiable?
5. Who should own the dry-run decision and the stop decision?

Recommended management question

Is the current workflow useful enough to justify a small, controlled dry-run of the future project-management, AI-limit, and feedback model?

Appendix A — Engineering evidence and exact version

This appendix records the main evidence without changing the plain-language management claim.

Evidence item	Recorded result
Product evidence version	5088dbf6e5fdeea2415b991c429de8e5229d7a7b
Repository checks	421 passed: 104 prompt optimizer, 31 workspace, 66 core, 16 catalog, 172 host, and 32 webview
Type checks	All six package checks passed
Schema check	No unexpected change
Build and package	Passed
Package contents	387 files; 684,130 bytes
Package SHA-256	954ce0 102 754b79f7b3503bfe0c9053ba30ae21ab95 177cc55d28c7643d2567b
Installed local flow	Passed for the recorded same-process journey
Narrow-screen review	360 × 640 review rendered without horizontal overflow
Exact-version CI	Verify DWI passed for the recorded implementation version

Restart evidence boundary

An earlier version passed the full restart, resume, and reset journey. The latest repeat was blocked before product checks because macOS automation could not see the consent window. The earlier result remains valid for its version; the latest version still needs its own fresh result.

Live external-AI quality boundary

One approved readiness call reported 14,531 tokens against an agreed 12,000 limit. The process stopped safely. Quality calibration and the controlled comparison did not start, and no retry was made.

Appendix B — Key risks and work still to do

Risk or unfinished work	Why it matters	What closes it
Guided answer step is incomplete	DWI can show a question but cannot finish the simple answer journey	Build and test answer, skip, safe use, and old-answer protection
Latest restart result is missing	Earlier evidence cannot automatically prove the latest version	Repeat the full journey on the same exact version
Live external-AI quality is unknown	Local checks cannot prove quality, cost, or time with a real service	Obtain a new approved test agreement and run controlled checks
Project-management data could become hidden authority	Urgency might improperly affect privacy, safety, or availability	Keep fields minimal, make the link optional, and stop safely on bad data
AI usage limit could be mistaken for permission	Priority might trigger unauthorized spend or data sharing	Enforce the control order and require explicit external-AI action
Feedback could become employee monitoring	Trust can be lost and activity may be mistaken for productivity	Exclude content and names, use minimum group sizes, and prohibit ranking
Usage estimates may be mistaken for bills	Management decisions could use inaccurate cost information	Label estimates, show uncertainty, and compare with observed use later
Activity counts may become the goal	More usage can hide poor outcomes or repeated attempts	Measure usefulness, correction, time, limits, and safe fallback together

Appendix C — Plain-English glossary

Term	Plain-English meaning
Project Meta Context	The short project summary the developer has reviewed and approved
Proof of concept	A working local demonstration used to test value and controls before wider use
Project-management system	The tool that holds planned tasks, priorities, delivery periods, and work status
Token	A unit used to measure AI input and output; it helps estimate usage, not necessarily the final bill
AI usage limit	A clear cap on how much AI input and output may be used for a task
Local rule-based draft	A draft prepared by DWI without calling an external AI service
External AI service	The separately configured AI system used only after an explicit disclosed action
Fallback	Continuing safely with the local draft when an external AI action fails or is unavailable
Outcome-only feedback	Simple result information that does not copy prompts, output, code, or project content
Controlled comparison	A fixed test used to check whether the external-AI improvement is genuinely better
Exact-version evidence	A test result that belongs only to the exact product version that produced it

Final boundary: current local capabilities and future ideas are intentionally separated. Future project-management, AI-limit, and management-feedback features require their own design, approval, implementation, privacy review, and evidence.